



KA997

Alumina (Al_2O_3) 99.7%

KEY PROPERTIES

- Excellent abrasion resistance
- Excellent dielectric properties
- High corrosion resistance
- Good thermal conductivity

APPLICATIONS

- Labware
- Furnace construction
- Abrasives industry
- Machinery parts
- Ballistic protection
- Chemical industry

MAIN PROPERTIES

Property	Value	Unit
Density	3.9	g/cm^3
Flexural Strength	350	MPa
Hardness	16	GPa
Fracture Toughness	3.5	$\text{MPa}\cdot\text{Vm}$
Young Modulus	370	GPa
Thermal Conductivity (100°C)	30	$\text{W/m}\cdot\text{K}$
CTE (20–1000°C)	8.6	$\times 10^{-6} \text{ K}^{-1}$

* All properties measured at 20°C unless otherwise stated.

** Measured in a wall thickness of 2 mm

OPERATING CONDITIONS

Max working temp (air)		1550	°C
Recomended temp		<1500	°C
Continuous temp		1400	°C

ELECTRICAL BEHAVIOR

Resistivity (25°C)		>1E14	$\Omega\cdot\text{cm}$
Dielectric strength		~15 kV/mm	

MATERIAL INFORMATION

Material type		Al_2O_3
Machinability		Medium
Structure		Dense ceramic
Purity		99.7%
Color		Ivory

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

The information provided is for guidance only and does not constitute a guarantee.

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